

# Ethereum explored

Learn how the second biggest cryptocurrency by market cap works.

Fidelity Viewpoints



## Key takeaways

Ethereum is a digital asset network, and its token is the second largest cryptocurrency by market capitalization behind bitcoin.

Its differences compared to the Bitcoin network primarily revolve around its programmability.

Those thinking about buying ethereum should consider unique risks before entering the market.

Bitcoin may be the most widely known cryptocurrency, but among crypto enthusiasts, ethereum commands just as much buzz. While there are a lot of similarities between the 2 largest cryptocurrencies, there are also some significant, fundamental differences. Here's what you need to know about ethereum and how it works.

## What is Ethereum?

Like the Bitcoin network, the Ethereum network is a digital asset network built on blockchain technology that offers a digital currency: Ethereum, typically abbreviated as "ETH."

However, one could argue that the 2 are more different than they are alike. For example, there is a finite supply of bitcoin, but a potentially unlimited supply of ETH. And while bitcoin was created solely to be used as a currency, the Ethereum network also allows other applications to be programmed on top of it, a bit like how unique smartphone apps can be built for smartphones.

Despite these differences, many crypto advocates don't view Bitcoin and Ethereum as an either-or scenario. Both are often viewed as beneficial to the crypto industry as a whole.

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The table below summarizes some other differences between the Bitcoin and Ethereum networks.

|                                             | Bitcoin network                                                      | Ethereum network                            |
|---------------------------------------------|----------------------------------------------------------------------|---------------------------------------------|
| Primary purpose                             | Decentralized monetary network                                       | A multipurpose platform                     |
| Founder(s)                                  | Unknown                                                              | Vitalik Buterin and others                  |
| Speed of improvement implementation         | Slower and deliberate                                                | Faster and more responsive to user demand   |
| Programmable or smart contracts?            | Extremely limited                                                    | Yes                                         |
| Ability to host multiple tokens?            | No (only bitcoin)                                                    | Yes                                         |
| Monetary or token issuance policy           | Fixed, pre programmed, and has never changed                         | Has changed and is expected to change again |
| Auditability (i.e., how many tokens exist?) | Easy to audit at any time (can be done with consumer-grade computer) | Can be audited, but may be more difficult   |
| Initial funding                             | None (all tokens have been mined through proof of work)              | Crowdsourcing and "pre-mined" tokens        |

## Who created Ethereum?

Ethereum was created by computer programmer and crypto enthusiast Vitalik Buterin. Buterin released the white paper for the network in 2013, and the first ethereum token was made available for sale in 2014.

Unlike Bitcoin, which was founded by the mysterious pseudonym Satoshi Nakamoto, Ethereum's founding team and current developers are known, and post regular updates on [ethereum.org](https://ethereum.org).

## How is Ethereum updated?

While Bitcoin's [blockchain ledger](#) is updated by "miners" through a system called [proof of work](#), Ethereum runs on a different system called [proof of stake](#).

In this system, Ethereum holders can choose to "stake" their coins. Think of staking like depositing money in a special account. Coins that are staked are locked in this account and can't be used for anything else unless you choose to withdraw them. For each group of transactions, the blockchain randomly chooses one person with staked cryptocurrency to update the ledger.

One way to think of this is that [proof of work](#) relies on competition, while [proof of stake](#) operates more like a lottery system.

## What is ETH's price, and how is it determined?

The [current price of ETH](#) can be seen on crypto exchanges and traditional trading platforms that offer crypto. This price can be influenced by many of the same factors that impact [bitcoin's price](#), including investor and user sentiment, changes in global economic data and world events, acceptance as a form of payment, network factors, and regulatory developments, among others.

## Tokens on the Ethereum network

Payments on the Ethereum network are made in ETH (its native token), and involve transaction fees commonly referred to as "gas fees." Gas fees are determined by how much activity is happening on the network. For example, if a lot of people are trying to make transactions at the same time, gas fees increase.

In addition to ETH, the Ethereum network allows developers to create unique tokens on top of its infrastructure. These non-ETH tokens on the Ethereum network can be fungible tokens (meaning they can be exchanged for other tokens) or [non-fungible tokens](#) (NFTs). Fungible tokens,

also known as ERC-20 tokens, are often used in the funding and development of decentralized applications on the Ethereum network.

## What is the connection between Ethereum and DeFi?

Proponents of Ethereum believe it's the ideal network for developing the decentralized finance (commonly referred to as "DeFi") industry.

DeFi aims to use blockchains to fulfill traditional finance functions, like payments, lending, borrowing, saving, trading, and insurance, without the need for intermediaries like banks and other financial institutions. It accomplishes this through smart contracts, which are digital contracts that can be programmed so that a predetermined action happens once certain requirements are met.

Ethereum was built to support smart contracts, and much of the DeFi industry is currently built using its infrastructure.

## What should I know about trading ethereum and taxes?

Sales or other dispositions of ETH (and other cryptocurrencies) are generally taxable events. Taxes on cryptocurrencies are similar in many respects to those of other capital assets like stocks and bonds—with short- and long-term holding period tax rules applying.

But beware that complexity can exist with cryptocurrency taxes, such as if you are actively trading and making many cryptocurrency trades. Also, there is no wash sale rule for cryptocurrencies. Consider consulting a tax professional if you have cryptocurrency tax questions.

## What does Ethereum's future look like?

The Ethereum network periodically undergoes major upgrades that aim to improve security, speed, token economics, and how it functions overall. For example, in 2022, the network initiated an event called The Merge, which transitioned it from proof of work to proof of stake. The development team has made upgrades nearly every year since its inception, and has more planned for the future.

In a nutshell, you can expect the network to continue to evolve.

## Risks of buying Ethereum: Is it secure?

There's no question that Ethereum is at the forefront of innovation in the crypto industry. But those thinking about buying ethereum should consider the following before jumping in:

- In general, crypto is highly volatile, and may be more susceptible to market manipulation than securities. The price of ethereum and other digital currencies has fluctuated unpredictably and drastically. You could experience significant and rapid losses as well as gains.
- Crypto (including ethereum) holders do not benefit from the same regulatory protections applicable to registered securities, and the future regulatory environment for crypto is currently uncertain. Crypto is not insured by the Federal Deposit Insurance Corporation (FDIC) or the Securities Investor Protection Corporation (SIPC).
- Digital currency such as ethereum is not currently legal tender. No law requires companies or individuals to accept ethereum as a form of payment. Instead, ethereum use is limited to businesses and individuals that are willing to accept it.
- Platforms that allow you to buy and sell ethereum may be unregulated, can be hacked, may stop operating, and can fail. In addition, like the platforms themselves, digital wallets can be hacked. As a result, consumers can—and have—lost money.
- Ethereum runs on proof of stake, which is a newer and relatively less proven system compared to proof of work. It could experience unforeseen challenges related to the proof of stake model down the road.
- Ethereum payments are designed to be irreversible, even in the event of fraud or theft. Once you complete a transaction, you cannot reverse it. Reversing a transaction depends solely on the willingness of the recipient to send back your ethereum.

In light of these risks, make sure you thoroughly understand how ethereum and crypto work before you buy. If you decide it's right for your portfolio, only buy with an amount you're willing to lose. This may help reduce the impact to your portfolio in the event of unforeseen circumstances.

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